



Course Code: CSE475
Course Title: Machine Learning

Section: 06

[Semester: Summer 2026]

Project Task: 01

Project Title: Explainable ML model for Skin-Cancer Detection

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Related Work Table:

Title	Dataset name and URL	Dataset description (samples, classes, images per class or split)	Method s name	Accura cy of the model	Researc h Questio ns	Pros and Cons	Citation
DSCC_Net: Multi-Class ification Deep Learning Models for Diagnosing of Skin Cancer Using Dermoscopic Images	Deep Learning Models for Diagnosing of Skin Cancer Using Dermoscopic Images ISIC 2020: challenge2020.isicarc hive.com HAM10000 URL: NR DermIS: dermis.net	4300 images; 4 classes. BCC 510, MEL 1686, MN 2007, SCC 97. After SMOTE Tomek: 8012 images. Train/validati on/test split: NR.	Custom DSCC_Net CNN; SMOTE Tomek; Grad-CA M. Compared with ResNet-15 2, VGG-16/1 9, MobileNet , EfficientN et-B0 and Inception-V3	Accuracy 94.17% Precision 94.28% Recall 93.76% F1 93.93% AUC 99.43%	Can a custom CNN classify MEL, BCC, SCC and MN better than common deep networks while addressing class imbalance?	Pros: • High four-class accuracy. • Imbalance handling and XAI. Cons: • Fair-toned images only. • Split and key hyperparam eters NR. • Some reported metrics conflict across sections.	M. Tahir et al., “DSCC_Net: MultiClassification Deep Learning Models for Diagnosing of Skin Cancer Using Dermoscopic Images,” Cancers, vol. 15, Art. 2179, 2023, doi: 10.3390/cancers15072179.
Deep Learning-Based Skin Diseases Classification using Smartphones	PAD-UFES-20 URL: NR Monkeypox Skin Lesion Dataset URL: NR	7 classes: ACK, BCC, MEL, NEV, SCC, SEK and MPX. Text reports 2298 PAD-UFES-20 + 102 MSLD images. Table reports MPX 1428, creating a conflict. Split: 80%/20%.	ImageNet transfer learning with seven lightweight CNNs. Selected ResNet-18 converted to TensorFlow Lite and deployed on Android. Training data augmentation used.	ResNet-18 : 74.27% Precision 76.78% Recall 71.40% F1 73.63% Jaccard 59.68% EfficientNetB3: 75.88%	Can a compact pretrained CNN perform seven-class skindisease classification, including MPX, in real time on smartphones?	Pros: • Working Android deployment . • Speed and FPS tested on four phones. Cons: • Lower accuracy than other reviewed models. • MPX count conflict and possible source bias. • No prospective clinical validation.	I. Oztel, G. Y. Oztel, and V. H. Sahin, “Deep Learning-Based Skin Diseases Classification using Smartphones,” Adv. Intell. Syst., vol. 5, Art. 2300211, 2023, doi: 10.1002/aisy.202300211.

Skin cancer classification using vision transformers and explainable artificial intelligence	ISIC: isic-archive.com Specific Kaggle subset; thirdparty URL omitted.	3297 images; 2 classes. Training: 2637 (1440 benign, 1197 malignant). Testing: 660 (360 benign, 300 malignant). SMOTE applied only to training data. Validation split: NR.	ViT-base/large; Swin-small/base/tiny; VGG19, ResNet18/50, DenseNet201 and MobileNetV2. ImageNet transfer learning. Grad-CAM, Grad-CAM++ and Score-CAM.	Best: ResNet50 88.8% Precision 86.9% Sensitivity 88.6% F1 87.8% Specificity 88.9% ViT-base/large: 88.6%	How do ViT, Swin and CNN models compare for benignversus-malignant classification, and can CAM-based XAI explain the best model?	Pros: • Direct CNN-transformer comparison. • Three XAI methods. Cons: • Binary task and small dataset. • XAI not quantitatively validated. • Some heatmaps focus on irrelevant regions. • No external validation.	G. H. Dagnaw, M. El Mouhtadi, and M. Mustapha, "Skin cancer classification using vision transformers and explainable artificial intelligence," J. Med. Artif. Intell., vol. 7, Art. 14, 2024, doi: 10.21037/jmai-24-6.
Enhancing Dermatological Diagnostics with EfficientNet: A Deep Learning Approach	Authors' clinical collection: private, URL NR ISIC 2019: gallery.isicarchive.com	8222 original images; 6 classes. MEL 1655, BCC 1811, BKL 1663, MN 1686, SCC 606, AK 801. After augmentation: 2144/class. Training 6578; validation + test 1644; separate split NR.	Customized ImageNet-pretrained EfficientNetB3 with batch normalization, 256-unit dense layer, dropout 0.45 and Softmax. Adamax, categorical crossentropy and augmentation.	4 classes: 95.4% validation; test 0.96 6 classes: 88.8% validation; test 0.89 AUC: 0.98-1.00 (4 classes); 0.93-1.00 (6 classes)	Can a customized EfficientNetB3 provide accurate and computationally efficient classification of four and six skinlesion categories?	Pros: • Strong four-class performance. • Class-wise metrics, ROC and confusion matrices. Cons: • Accuracy drops with AK and SCC. • SCC recall 0.54, F1 0.61. • Separate validation/test split NR. • No independent external validation.	I. Manole, A.-I. Butacu, R. N. Bejan, and G.-S. Tiplica, "Enhancing Dermatological Diagnostics with EfficientNet: A Deep Learning Approach," Bioengineering, vol. 11, Art. 810, 2024, doi: 10.3390/bioengineering11080810.

EfficientSkinDis: An EfficientNet-based classification model for a large manually curated dataset of 31 skin diseases	Atlas Dermatology: atlasdermatologico.com.br ISIC: isic-archive.com Released dataset: Google Drive link in paper	4910 images; 31 classes; original class sizes 80-478. Final split: 3916 training and 994 testing. Augmented training count conflicts: 44,918 in Table 2 vs 37,551 in text.	Transfer-learned EfficientNet-B0- B6, ResNet-50 /101/152 and VGG-16/19. Best: fine-tuned EfficientNet-B2. Six augmentations, 10-fold crossvalidation and Streamlit deployment.	EfficientNet-B2: 87.15% Precision 0.87 Recall 0.87 F1 0.87 Mean AUC 0.91 10-fold range: 0.8511-0.8715	Which transfer-learned CNN performs best for 31 diseases, and how do split choice and augmentation affect performance?	Pros: • Widest disease coverage. • Twelve-model comparison and 10-fold validation. • Public web demonstration. Cons: • Source and acquisition bias possible. • Patient-level split and label verification NR. • Model selection relied on test accuracy. • Augmented counts conflict.	A. Rafay and W. Hussain, "EfficientSkinDis: An EfficientNetbased classification model for a large manually curated dataset of 31 skin diseases," Biomed. Signal Process. Control, vol. 85, Art. 104869, 2023, doi: 10.1016/j.bspc.2023.104869.
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